



CS-467- Software Project Management

Lecture No. 5

Software Project Scheduling

List of Topic to be Discuss in this lecture

- Project Schedule Management
- Plan Schedule Management Process
- Define Activities
- Sequence Activities
- Precedence Diagram Method /Project Network Diagram
- Estimate Activity Resource
- Estimate Activity Duration
- Control Schedule Process
- Steps to Develop a Project Schedule

Knowledge Area -Schedule Management

Knowledge Area	Process				
	Initiating	Planning	Executing	Monitoring & Control	Closing
Schedule Management		Plan Schedule Management Define Activities Sequence activities Estimate activity resource Estimate activity duration Develop Schedule		Control Schedule	

Project Schedule Management

- Scheduling in project management is the listing of **activities**, **deliverables**, and **milestones** within a project.
- A schedule also usually includes a planned **start and finish date**, **duration**, and **resources** assigned to each activity.
- Effective project scheduling is a **critical component** of successful time management.

Plan Schedule Management Process

Plan schedule management is the process of how to **plan**, **manage** and **control schedule**.

It involves information such as

- How to determine measures of performance,
- How and when to capture data for measuring,
- How to use the data to keep the project on track with schedule baselines,
- What actions should be taken when variances occur?

Define Activities Process

Define activities lets project managers use the **Work Breakdown Structure (WBS)** to establish start the assignments that are essential to finish on time. Thus ensuring they aware of what activities have to be included in the plan.

Sequence Activities Process

- Sequence activities help to determine the **relationship** between the **project activities**. The ordering of the tasks and **identification of dependencies** (Finish to Start, Finish to Finish, Start to Start, Start to Finish) is the next step.
- The activities and milestones will be **organized** according to the **order** in which the **work will be done** on the project.
- The **output of Sequence Activities Process is Project Schedule Network Diagram.**

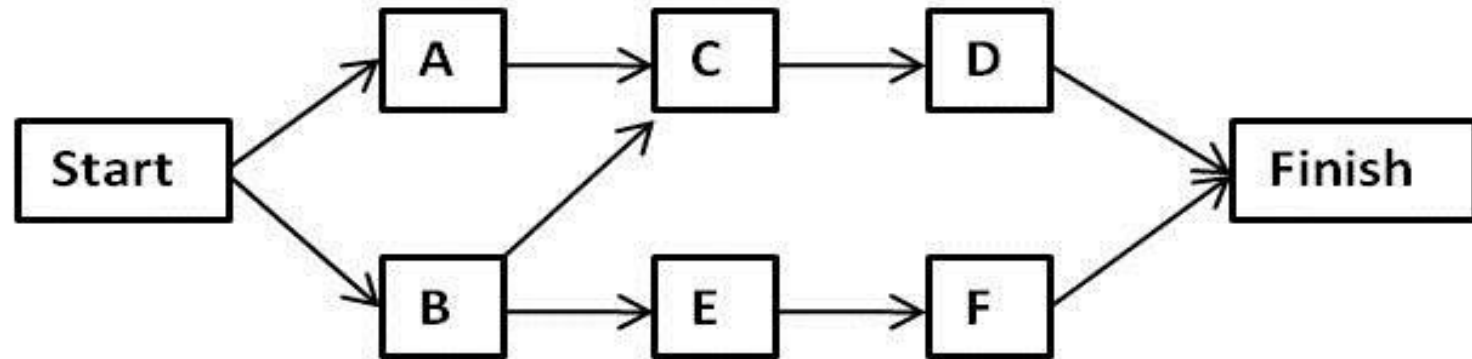
Precedence Diagram Method /Project Network Diagram

The **Precedence Diagramming Method** (PDM) consists of rectangles known as nodes, and the project activities are shown in these boxes.

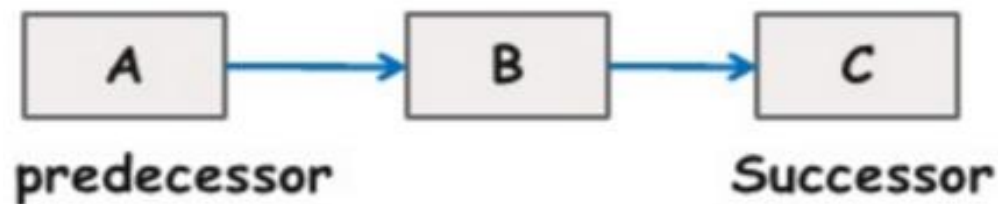
These rectangular boxes are connected to each other through an arrow to show the dependencies; therefore, these diagrams are also known as the **Activity on Node** (AON) diagrams.

The main benefit of Precedence Diagramming Method (PDM) is that it shows the **activity dependencies** and it also an important communication tool for stakeholders.

Precedence Diagram Method /Project Network Diagram



Precedence Diagram Method (PDM)



Types of Dependencies

Finish-to-start: The predecessor activity must finish before the successor activity can start. This is the most common relationship type. *Example:* The Web server must be running before the SQL database can connect.

Finish-to-finish: Two activities must finish at the same time. The two activities don't have to begin at once, but they must end at the same time. *Example:* The user acceptability testing and the third-party reviews of the build must complete at the same time.

Start-to-start: Two activities must begin at the same time. The two activities may be of varying duration, so they don't have to end at the same time, but they have to start at the same time. *Example:* The final build of the software and the creation of the online help system should start at the same time.

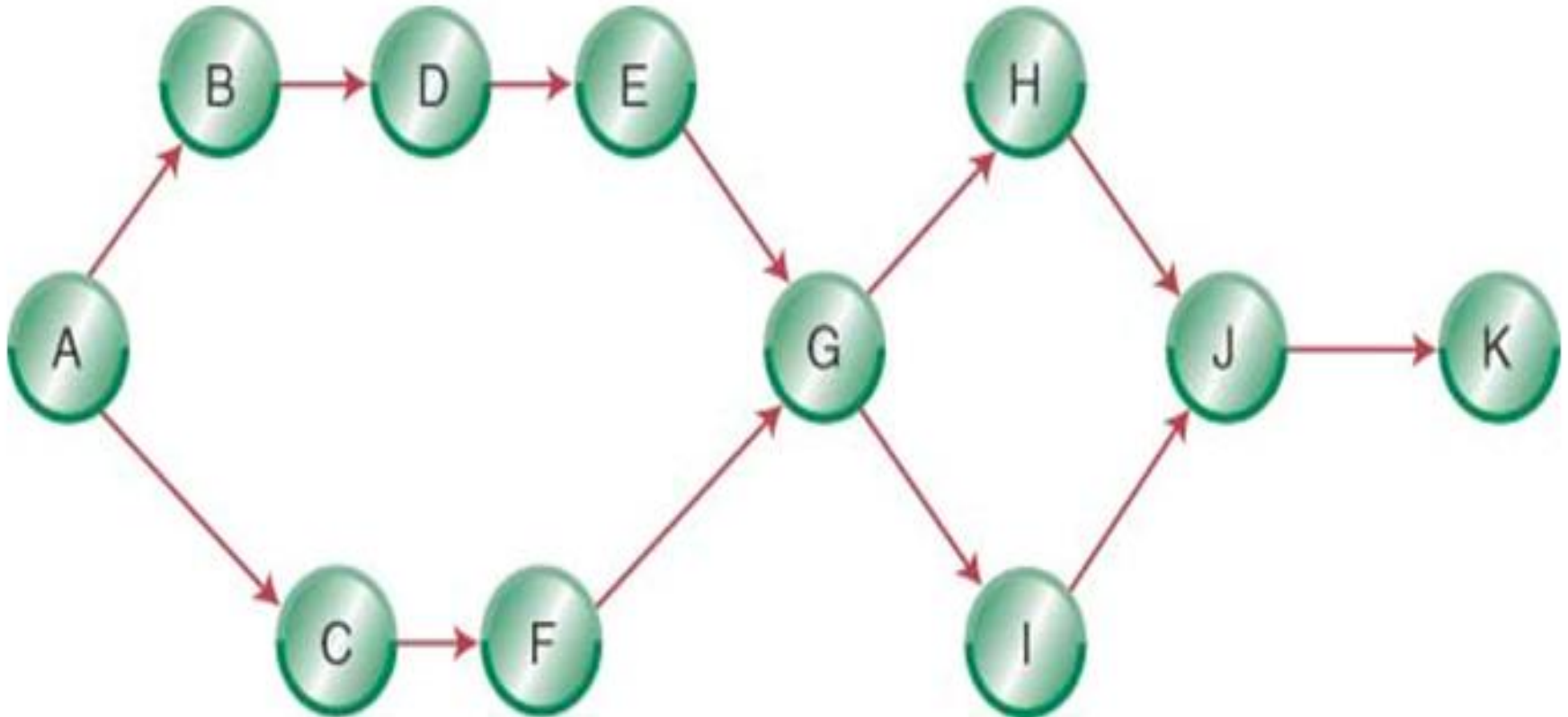
Start-to-finish: The predecessor activity must start in order for the successor activity to finish. This is the most unusual of all the relationship types and is rarely used. You might run across it if you do *just-in-time* scheduling or manufacturing. *Example:* The software testing can't be completed until the quality control department reviews the testing results.

Project Network Diagram - Problem

Activity	Description	Immediate Predecessor	Duration (weeks)
A	Develop product specifications	None	4
B	Design manufacturing process	A	6
C	Source & purchase materials	A	3
D	Source & purchase tooling & equipment	B	6
E	Receive & install tooling & equipment	D	14
F	Receive materials	C	5
G	Pilot production run	E & F	2
H	Evaluate product design	G	2
I	Evaluate process performance	G	3
J	Write documentation report	H & I	4
K	Transition to manufacturing	J	2

Project Network Diagram - Solution

Step 1- Draw the Network Diagram

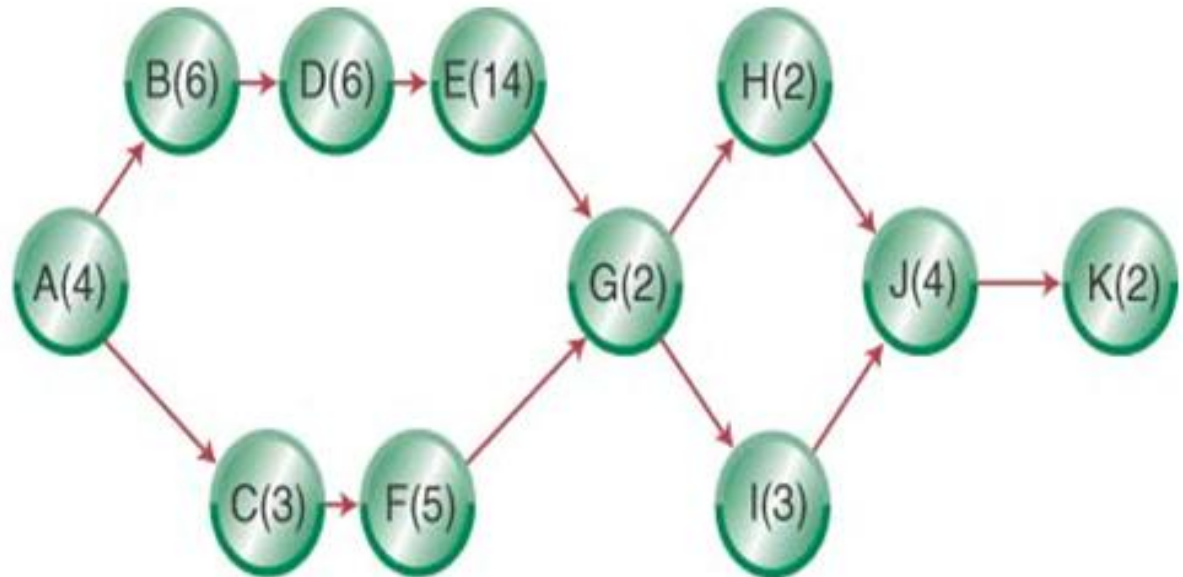


Project Network Diagram - Solution

Step 2 - Add Deterministic Time Estimates and Connected Paths

Connected paths

1. A, B, D, E, G, H, J, K
2. A, B, D, E, G, I, J, K
3. A, C, F, G, H, J, K
4. A, C, F, G, I, J, K



Step 3 - Calculate the Project Completion Times

Paths	Path duration
ABDEGHJK	40
ABDEGIJK	41
ACFGHJK	22
ACFGIJK	23

Project Network Diagram - Solution

Step 4 – Find the critical path

- The longest path (**ABDEGIJK**) limits the project's duration (project cannot finish in less time than its longest path).
- Hence Path 2: **ABDEGIJK** is the project's **Critical path**.

Point to Remember:

- ✓ Use the **critical path method** to avoid big problems.
- ✓ The **critical path method** is an important tool for keeping your projects on track. Every network diagram has something called the **critical path**.
- ✓ A *delay in any one of the critical path activities* will cause the **entire project to be delayed**.

Estimate Activity Durations Process

Once you're done with Estimate Activity Resources, you've got everything you need to figure out how long each activity will take. That's done in a process called **Estimate Activity Durations**.

You'll use these five tools and techniques to create the most accurate estimates.

- **Expert estimation:** Expert estimation means that an expert estimates how much effort a project requires.
- **Analogous:** Estimation of a task using a previous or similar project. For example, the flooring in the previous house take 3 months therefore this one should be Atleast 3 months.

Estimate Activity Durations Process

- **Parametric:** It is also called **Problem Based Estimation**. Parametric estimation methods use a model or algorithm that takes as input some aspects of the project (such as the required functionality and the quality that is expected). The model (a formula) or algorithm (computational steps) then produce an estimate based on those inputs alone. For example, if constructing a ten foot wall took one day, then how long will it take to build a hundred foot of wall? —You will multiply the time taken to build ten foot of wall in the old project by ten.
- **Project Evaluation and Review Technique (PERT) estimation:** Three point estimating is a technique which utilizes an optimistic and pessimistic estimate to determine the ideal estimate value for a project task.
- **Reserve analysis** means adding extra time to the schedule (called a *contingency reserve* or a *buffer*) to account for extra risk.

PERT (Program Evaluation Review Technique)

One approach to creating an accurate time estimate is something called PERT. PERT means program evaluation and review technique; it's an estimating approach that requires **three estimates** for every activity to account for uncertainty:

There are two generally accepted formulas for three point estimates:

- Triangular Distribution
- Beta Distribution

Three Point Estimation

Triangular Distribution

The simplest three point estimate is the simple average of the three values (known as the triangular distribution):

$$E = (O + M + P) / 3$$

Where:

- **E** = Estimated cost
- **O** = Optimistic: The best-case scenario
- **M** = Most likely: The most-likely time
- **P** = Pessimistic: The worst that can happen

PERT Beta Distribution

The beta distribution places the final estimate closer to the most likely value:

$$E = (O + 4M + P) / 6$$

Three Point Estimation - Numerical

Question:

A project manager used data from past projects to come up with an estimate for an upcoming software system replacement project. She felt confident about 25-day duration, but also noted that adding an extra resource could bring the schedule down to 10 days. The test team felt that the complexity of some completely new features would add additional test cases, adding a few weeks for a 40-day estimate. **Find the Optimistic, Most Likely, Pessimistic and Expected durations respectively by using triangular distribution method.**

Solution:

Optimistic duration = 10d; Most likely duration = 25d; Pessimistic duration = 40d

Expected duration = $40 + 10 + 4(25) / 6$

Expected duration = 25days

Develop Schedule Process

- The **Develop Schedule process** is the core of time management.
- It's the process where you put it all together—where you take everything you've done so far and combine it into one **final schedule for the whole project**.
- The **Outputs** of Develop Schedule process is **Project Schedule** or **Work Plan** or **Work Schedule**.

Control Schedule Process

As the project work is happening, you can always discover new information that makes you re-evaluate your plan, and use the **control schedule process** to make the changes.

Control Schedule tools and techniques:

The tools and techniques for Control Schedule are all about figuring out where your project schedule stands. By comparing your actual project to the schedule you laid out in the baseline and looking at how people are performing, you can figure out how to handle every schedule change.

Performance reviews:

There are two important calculations called **schedule variance (SV)** and **schedule performance index (SPI)** that give you valuable information about how your project is doing.

Steps to Develop a Project Schedule

Steps:

1. Determine the **tasks** to be placed in the schedule
2. Determine the **relationships** between the tasks
3. **Assign** each task to specific staff
4. Estimate the **amount of effort** required for each task
5. Consider the **other variables** that go into building the schedule
6. Build a time reserve into the schedule for **contingencies** and **unforeseen events**
7. Identify the project's **critical path**
8. Check to see if the staff is over-allocated
9. Repeat steps 3 and 5-8 until a baseline is established.
10. Place the schedule information in a **Gantt chart**